

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1
 Duration: 30 minutes

Semester: Spring 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1. Given that Computer has a MIPS of 5, and it requires 5 seconds to execute a specific program, CPI is 2. **Determine** the number of total clock cycles for that program. [2]

Answer: $MIPS = 5$; 5M instructions run in 1s.

$$\begin{array}{l|l}
 1s - 5M \text{ ins.} & \text{Clock Cycle} = \text{Instruction Count} \times \text{CPI} \\
 \therefore 5s - (5 \times 5) \text{ s} & = 25M \times 2 \\
 = 25M \text{ ins.} & = 50M = 50 \times 10^6 \text{ (Ans)}
 \end{array}$$

2. Computer A has a MIPS of 10. Tasks mentioned in the table below will be run in computer A.

	Name of the Task	Number of Instruction Count	Clock Rate (GHz)	Reference Time (s)	CPU time	SPEC Ratio
1	A	1×10^9	10	2	100s	0.02
2	B	2.5×10^9	12	1.5	250s	6×10^{-3}
3	C	1.5×10^9	25	2.9	150s	0.019

- a) Find the spec ratio for each of the Tasks. [4.5]

Answer: $MIPS = 10$; 10M instructions run in 1s.

$$\begin{array}{l|l}
 10M - 1s & \\
 1 - \frac{1}{10M} & \\
 \therefore 1 \times 10^9 - \frac{1 \times 10^9}{10 \times 10^6} & \\
 = 100s & \\
 \hline
 2.5 \times 10^9 - \frac{2.5 \times 10^9}{10 \times 10^6} = 250 & \text{SPEC Ratio} = \frac{\text{Reference Time}}{\text{CPU time}} \\
 1.5 \times 10^9 - \frac{1.5 \times 10^9}{10 \times 10^6} = 150 &
 \end{array}$$

- b) Find the geometric mean and mention if computer A performs better than the reference computer. Also, Justify your answer [2 + 1 + 0.5]

Answer:

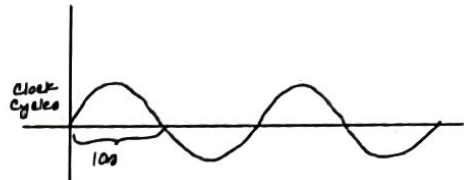
$$G.M. = \sqrt[3]{0.02 \times 6 \times 10^{-3} \times 0.019} = 0.013 \text{ (Ans)}$$

$G.M. < 1$. So, Computer A is slower than the reference computer / Computer A does not perform better than reference computer.

3.

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ADD X20, X21, X22
ADD X20, X21, X21
SUB X20, X21, X22
SUB X20, X21, X22
ADD X20, X21, X22
ADD X20, X21, X22
LD X20, 10(X21)
ADD X20, X21, X22
SD X21, 10(X20)
    
```



CPI for Add, Sub, LD, SD instructions are 3, 2, 1, 4.

- Find the number of instructions in the above code. [1]
- Find the average CPI. [1]
- Find the clock cycle time. [1]
- Find the time to execute this program. [2]

a) 9

$$\begin{aligned}
 \text{b) Avg. CPI} &= \frac{5 \times 3 + 2 \times 2 + 1 \times 1 + 1 \times 4}{9} \\
 &= 2.67
 \end{aligned}$$

	IC	CPI
Add =	5	3
Sub =	2	2
LD =	1	1
SD =	1	4

c) Clock cycle = time taken to complete 1 clock cycle

$$= 10 + 10 = 20\text{s}$$

d) Execution time = Instruction count $\times$ Avg. CPI $\times$ Clock cycle

$$\begin{aligned}
 &= 9 \times 2.67 \times 20 \\
 &= 480.6\text{ s (Ans)}
 \end{aligned}$$